🏆 Infosys SE Elite Crack Vault — 03 Digit Manipulation and Digital Root Axioms Elite

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Curation Standard: 100% Pure SE Role Specific** (All SP/DSE Upper-Level & Practice Variants Purged)
> **Total Curated Questions in Topic: 10 Questions**
> **Question Breakdown:** 🔴 `3 Must-Solve` | 🟡 `7 Core SE Level`
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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Question 1: Matrix Number Puzzle 46 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
21	29	25
27	17	22
26	20	?

Options

- [x] **A)** 23
- [ ] **B)** 28
- [ ] **C)** 18
- [ ] **D)** 38

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option A) 23**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (26 + 20) ÷ 2 = 23.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 23$.

4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 23.

Distractor & Elimination Analysis

- **Option A (23):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (28):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (38):** Incorrect. Violates the established matrix row/column transformation rule.

Question 2: Matrix Number Puzzle 47 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	25	16
30	6	18
13	13	?

Options

- [ ] **A)** 10
- [x] **B)** 13
- [ ] **C)** 25
- [ ] **D)** 31

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 13**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (13 + 13) ÷ 2 = 13.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 13$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 13.

Distractor & Elimination Analysis

- **Option A (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (31):** Incorrect. Violates the established matrix row/column transformation rule.

Question 3: Matrix Number Puzzle 48 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
10	24	17
24	26	25
26	24	?

Options

- [ ] **A)** 31
- [ ] **B)** 21
- [x] **C)** 25
- [ ] **D)** 15

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 25**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (26 + 24) ÷ 2 = 25.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 25$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 25.

Distractor & Elimination Analysis

- **Option A (31):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (25):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

- **Option D (15):** Incorrect. Violates the established matrix row/column transformation rule.

Question 4: Matrix Number Puzzle 49 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
27	29	28
19	23	21
4	26	?

Options

- [x] **A)** 15
- [ ] **B)** 12
- [ ] **C)** 20
- [ ] **D)** 5

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option A) 15**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (4 + 26) ÷ 2 = 15.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 15$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (20):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (5):** Incorrect. Violates the established matrix row/column transformation rule.

Question 5: Matrix Number Puzzle 50 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
10	20	15
25	25	25
24	24	?

Options

- ☐ **A)** 34
- ☐ **B)** 30
- ☑ **C)** 24
- ☐ **D)** 19

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 24**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = average of first and second. So ? = (24 + 24) ÷ 2 = 24.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 24$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 24.

Distractor & Elimination Analysis

- **Option A (34):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (24):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (19):** Incorrect. Violates the established matrix row/column transformation rule.

Question 6: Matrix Number Puzzle 51 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
11	9	18
7	9	9
6	4	?

Options

- ☐ **A)** 3
- ☐ **B)** 18
- ☐ **C)** 2
- ☑ **D)** 6

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 6**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 6×4=24; digit sum = 6.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 6$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 6.

Distractor & Elimination Analysis

- **Option A (3):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (2):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (6):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 7: Matrix Number Puzzle 52 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	7	4
6	7	6
20	2	?

Options

- ☐ **A)** 19
- ☑ **B)** 4
- ☐ **C)** 9
- ☐ **D)** 22

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 4**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 20×2=40; digit sum = 4.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 4$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4.

Distractor & Elimination Analysis

- **Option A (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (22):** Incorrect. Violates the established matrix row/column transformation rule.

Question 8: Matrix Number Puzzle 53 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	7	12
12	2	6
18	6	?

Options

- [ ] **A)** 24
- [ ] **B)** 13
- [ ] **C)** 17
- [x] **D)** 9

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 9**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 18×6=108; digit sum = 9.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 9$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9.

Distractor & Elimination Analysis

- **Option A (24):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 9: Matrix Number Puzzle 54 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
18	7	9
15	3	9
5	2	?

Options

- ☐ **A)** 6
- ☑ **B)** 1
- ☐ **C)** 11
- ☐ **D)** 9

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 1**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 5×2=10; digit sum = 1.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 1$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 1.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (1):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (9):** Incorrect. Violates the established matrix row/column transformation rule.

Question 10: Matrix Number Puzzle 55 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 04: Digit Manipulation and Digital Root Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	6	15
7	9	9
14	4	?

Options

- ☐ **A)** 16
- ☑ **B)** 11
- ☐ **C)** 26
- ☐ **D)** 19

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 11**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row gives the digit sum of (first × second). For the last row: 14×4=56; digit sum = 11.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 11$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 11.

Distractor & Elimination Analysis

- **Option A (16):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (11):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (26):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (19):** Incorrect. Violates the established matrix row/column transformation rule.
